

```
from google.colab import files
```

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Revision history

Notebook info

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/csv) - 11960454 bytes, last modified: 6/30/2026 - 100% done
 fb6.csv to 617ec7a0-b7f1-423e-b810-23f59803ffb6.csv

SATISFACTION PREDICTION

whether an airline passenger is satisfied or dissatisfied based on travel characteristics
 the key factors influencing customer satisfaction and provides recommendations for

```
train_test_split  

encoder  

cRegression  

trix  

re  

ore  

on_report
```

Double-click (or enter) to edit

```
df = pd.read_csv("617ec7a0-b7f1-423e-b810-23f59803ffb6.csv")
```

```
df.head()
```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	...	Online support	Ease of Online booking
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0	0	0	2	...	2	
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0	0	0	3	...	2	
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0	0	0	3	...	2	
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0	0	0	3	...	3	
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0	0	0	3	...	4	

5 rows × 22 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  

RangeIndex: 129880 entries, 0 to 129879  

Data columns (total 22 columns):  

#   Column                                Non-Null Count  Dtype  

---  -  

0   satisfaction                          129880 non-null object  

1   Customer Type                        129880 non-null object  

2   Age                                  129880 non-null int64  

3   Type of Travel                       129880 non-null object  

4   Class                                129880 non-null object  

5   Flight Distance                      129880 non-null int64  

6   Seat comfort                         129880 non-null int64  

7   Departure/Arrival time convenient    129880 non-null int64  

8   Food and drink                       129880 non-null int64  

9   Gate location                        129880 non-null int64  

10  Inflight wifi service                 129880 non-null int64  

11  Inflight entertainment                129880 non-null int64
```

```

12 Online support          129880 non-null int64
13 Ease of Online booking  129880 non-null int64
14 On-board service        129880 non-null int64
15 Leg room service        129880 non-null int64
16 Baggage handling        129880 non-null int64
17 Checkin service         129880 non-null int64
18 Cleanliness             129880 non-null int64
19 Online boarding         129880 non-null int64
20 Departure Delay in Minutes 129880 non-null int64
21 Arrival Delay in Minutes 129487 non-null float64
dtypes: float64(1), int64(17), object(4)
memory usage: 21.8+ MB

```

```
df.describe()
```

	Age	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	Inflight wifi service	Inflight entertainment
count	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000
mean	39.427957	1981.409055	2.838597	2.990645	2.851994	2.990422	3.249130	3.383000
std	15.119360	1027.115606	1.392983	1.527224	1.443729	1.305970	1.318818	1.346000
min	7.000000	50.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	27.000000	1359.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000
50%	40.000000	1925.000000	3.000000	3.000000	3.000000	3.000000	3.000000	4.000000
75%	51.000000	2544.000000	4.000000	4.000000	4.000000	4.000000	4.000000	4.000000
max	85.000000	6951.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000

```
df.isnull().sum()
```

	0
satisfaction	0
Customer Type	0
Age	0
Type of Travel	0
Class	0
Flight Distance	0
Seat comfort	0
Departure/Arrival time convenient	0
Food and drink	0
Gate location	0
Inflight wifi service	0
Inflight entertainment	0
Online support	0
Ease of Online booking	0
On-board service	0
Leg room service	0
Baggage handling	0
Checkin service	0
Cleanliness	0
Online boarding	0
Departure Delay in Minutes	0
Arrival Delay in Minutes	393

```
dtype: int64
```

```
df = df.dropna()
```

```
df["satisfaction"].value_counts()
```

```

count
satisfaction
satisfied    70882
dissatisfied  58605

dtype: int64

```

```

encoder = LabelEncoder()

categorical_columns = [
    "satisfaction",
    "Customer Type",
    "Type of Travel",
    "Class"
]
for col in categorical_columns:
    df[col] = encoder.fit_transform(df[col])

df.head()

```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	...	Online support	Ease of Online booking
0	1	0	65	1	1	265	0	0	0	2	...	2	3
1	1	0	47	1	0	2464	0	0	0	3	...	2	3
2	1	0	15	1	1	2138	0	0	0	3	...	2	2
3	1	0	60	1	1	623	0	0	0	3	...	3	1
4	1	0	70	1	1	354	0	0	0	3	...	4	2

5 rows × 22 columns

```

x = df.drop("satisfaction", axis=1)
y = df["satisfaction"]

```

```

from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.2,
    random_state=42
)

```

```

model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)

```

/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```

n_iter_i = _check_optimize_result(

```

```

    LogisticRegression

```

```

    LogisticRegression(max_iter=1000)

```

```

coef = pd.DataFrame({
    "Feature": x.columns,
    "coefficient": model.coef_[0]
})

print(coef.sort_values(by="coefficient", ascending=False))

```

	Feature	coefficient
10	Inflight entertainment	0.589606
12	Ease of Online booking	0.540878
5	Seat comfort	0.370577
16	Checkin service	0.282411
14	Leg room service	0.122274

13	On-board service	0.115758
8	Gate location	0.058037
18	Online boarding	0.031189
15	Baggage handling	0.004079
19	Departure Delay in Minutes	0.001000
4	Flight Distance	-0.000416
20	Arrival Delay in Minutes	-0.005015
1	Age	-0.023595
11	Online support	-0.058846
17	Cleanliness	-0.067522
9	Inflight wifi service	-0.196848
6	Departure/Arrival time convenient	-0.205294
7	Food and drink	-0.275007
2	Type of Travel	-0.480478
3	Class	-1.066137
0	Customer Type	-1.910779

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MODEL COEFFICIENT INTERPRETATION

1. Positive coefficients increase the probability that a passenger is satisfied.

* Inflight Entertainment (0.5896) - The strongest positive predictor of customer satisfaction.

* Ease of Online Booking (0.5409) - Customers who find booking easy are more likely to be satisfied.

* Seat Comfort (0.3706) - Comfortable seating significantly improves satisfaction.

* Check-in Service (0.2824) - Efficient check-in contributes positively to customer experience.

* Leg Room Service (0.1223) and On-board service (0.1158) also have positive effects.

2. Negative coefficients decrease the probability of satisfaction.

* Customer Type (-1.9108) - This variable has the greatest negative impact on satisfaction.

* Class (-1.0661) - Passenger class is associated with lower satisfaction for the encoded category.

* Type of Travel (-0.4805) - Certain travel types are linked with lower satisfaction.

* Food and Drink (-0.2750) and Departure/Arrival Time Convenience (-0.2053) also reduce the likelihood of customer satisfaction.

3. Feature with the largest absolute coefficients have the strongest influence on prediction.

Overall, the coefficients show that improving the quality of in-flight services and making the booking process easier are likely to increase customer satisfaction.

MODEL COEFFICIENT INTERPRETATION

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```
from sklearn.metrics import f1_score

print("f1-score:", f1_score(y_test, y_pred))

f1-score: 0.8323747777893966
```

MODEL EVALUATION

- Accuracy: (0.8143099853270522)
- Precision: (0.8171366000547495)
- Recall: (0.8481920863820416)
- f1-score: (0.8323747777893966)

```
y_pred = model.predict(x_test)
```

```
cm = confusion_matrix(y_test, y_pred)

print(cm)
```

```
[[ 9149 2672]
 [ 2137 11940]]
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))

print("Precision:", precision_score(y_test, y_pred))

print("Recall:", recall_score(y_test, y_pred))
```

```
Accuracy: 0.8143099853270522
Precision: 0.8171366000547495
Recall: 0.8481920863820416
```

```
print(classification_report(y_test, y_pred))
```

```

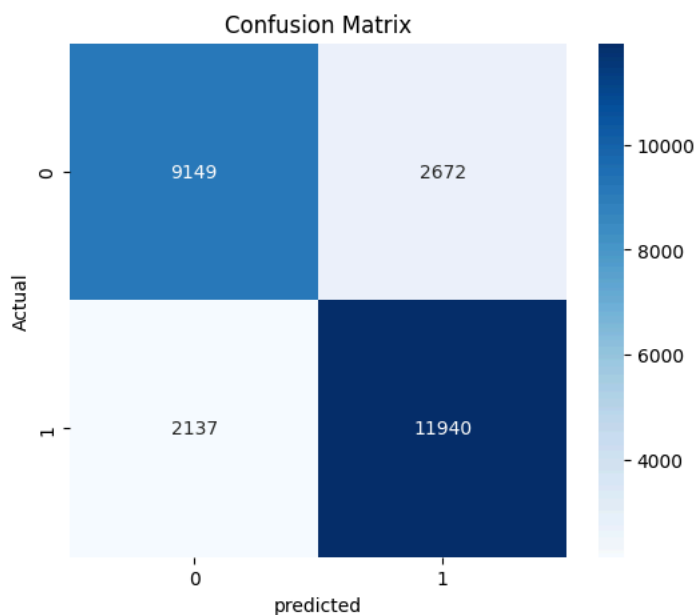
              precision    recall  f1-score   support

     0       0.81         0.77         0.79        11821
     1       0.82         0.85         0.83        14077

 accuracy          0.81         0.81         0.81        25898
 macro avg         0.81         0.81         0.81        25898
 weighted avg      0.81         0.81         0.81        25898
```

```
plt.figure(figsize=(6,5))
sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues')

plt.xlabel("predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



INTERPRETATION & EVALUATION

MODEL PERFORMANCE The Logistic Regression model achieved an accuracy of 81.43%, indicating that it correctly classified approximately 81 out of every 100 passengers.

The model achieved a precision of 81.71%, meaning that when the model predicts a passenger is satisfied, it is correct about 82% of the time.

The recall score of 84.82% indicates that the model successfully identified approximately 85% of all satisfied passengers.

Overall, these metrics demonstrate that the Logistic Regression model performs well and is suitable for predicting airline customer satisfaction.

CONFUSION MATRIX INTERPRETATION The confusion matrix shows the following results:

*True Negatives (TN): 9,149 passengers were correctly classified as not satisfied. *False Positive (FP): 2,672 passengers were incorrectly classified as satisfied. *False Negatives (FN): 2,137 satisfied passengers were incorrectly classified as not satisfied. *True Positives (TP):

11,940 satisfied passengers were correctly classified.

The relatively high number of true positives and true negatives indicates that the model effectively distinguishes between satisfied and dissatisfied passengers.

BUSINESS RECOMMENDATION

The Logistic Regression model demonstrates strong predictive performance with an accuracy of 81.43%, making it a valuable tool for understanding customer satisfaction.

Based on the model results, the airline should:

- Focus on improving service features that most strongly influence customer satisfaction, such as in-flight Wi-fi, seat comfort, online boarding, cleanliness, and in-flight entertainment.
- Use the model to identify passengers who are likely to be dissatisfied and provide targeted service improvements before future flights.
- Prioritize investments in customer experience initiatives that increase passenger satisfaction and loyalty.
- Continuously monitor customer feedback and retain the model with new data to improve prediction accuracy over time.

CONCLUSION

The Logistic Regression model successfully predicts airline customer satisfaction with good performance, achieving an accuracy of 81.43%, precision of 81.71%, and recall of 84.82%.

The results indicate that the model is effective for identifying satisfied and dissatisfied passengers. The airline can use these predictions to improve customer experience, strengthen customer retention, and make more informed business decisions regarding service quality.

LIMITATIONS & FUTURE WORK

Although the model performs well, there are opportunities for improvement:

- Additional feature engineering may improve prediction performance.
- Other classification algorithms, such as Random Forest, XGBoost, or support Vector Machines, could be compared with Logistic Regression.
- Future models should be retained using updated customer survey data to maintain prediction accuracy over time.